BerkeleyHaas

XMBA 201A

Economic Analysis for Business Decisions

Price Discrimination & Competitive Market Dynamics

Lecture 6

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Recap

- Optimal pricing: MR = MC (and check FC against shutdown)
- Price discrimination (PD) reduces CS and DWL, increasing profits
- In optimal Group-PD, MR's must be equalized, and they will be equal to MC when there is no capacity constraint.
- But these transfer surplus to the monopolist so some consumers may be worse off, while others might benefit.
- Two-part tariffs also reduce deadweight loss and increase profits
- Market segmentation (quantity discounts) also help reduce deadweight loss and consumer surplus to increase profits.

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Case: Brand X

1. Using the data, can you pass judgement on the pricing strategy of the Brand X manufacturer?
2. What pricing strategy would you recommend given the data?
3. What concerns should you have in answering (2) above?

Case: Brand X

- To figure the optimal price of Chemical Z, we need to know
  - Demand curve for Chemical Z
  - Brand X marginal cost for producing it
- Data:
  - Brand X has priced at "unit cost of raw materials" + 20% (This unit cost includes overhead costs!)
  - Chemical Z demand curve is said to be stable over time
  - Marginal Cost depends on the price of raw materials and labor

    → Cost+ pricing is not the right way to go!
  - MC varies from quarter to quarter

    → Optimal price varies from quarter to quarter

Brand X: Calculating MC

- MC for a gallon of chemical Z is the sum of:

  Cost of variable labor/gallon + Cost of raw materials/gallon
- This marginal cost is constant (independent of quantity) even though it varies over time
  - Cost of variable labor/gallon = (hourly wage) × (20/10000)
    - Hourly wage in quarter 40 = $29.19
    - Hourly wage in quarter t = $29.19/(1.002^{40-t})
- See column G in *chemical_plant_answers.xls*

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Brand X: Calculating MC

- Cost of raw materials/gallon
  - The old pricing rule stated that Price = (cost /gallon) × 1.2
  - Data: "raw materials as percentage of unit cost" = q (64%-72% in xls sheet)

    → (cost/gallon) = Price/1.2

    → (cost of raw materials /gallon) = q × (Price/1.2)
- See column H in *chemical_plant_answers.xls*

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Brand X: Estimating Demand

- Assume linear demand: $p(q) = a - bq$, or $q(p) = k - mq$ ($k = \frac{a}{b}, m = \frac{1}{b}$)
- Run a regression, say $q(p) = k - mq$ (MBA200S: Data & Decisions!)
- Resulting regression line is

$$q_t = 4,982,786 - 245,065p_t$$

- To then calculate $p(q)$:

$$p_t = 20.33 - 4.08\times10^{-6}q$$
$$(a = 20.33; b = 4.08\times10^{-6}\ or\ 0.00000408)$$

Brand X: Optimal pricing

- Recall that for linear demand: MR(q) = a – 2b × q, so:
  - $MR(q_t) = 20.33 - 8.16\times10^{-6}q$
- Optimal choice of q_t : MR(q_t) = MC(q_t)

  $a - 2bq_t = MC(q_t)$ (this is how we find optimal quantity)

  → solve for quantity: $q_t = \frac{a - MC(q_t)}{2b}$

  → solve for price using demand:

  $p_t = a - bq_t = a - b\left(\frac{a}{2b} - \frac{MC(q_t)}{2b}\right) = \frac{a}{2} + \frac{MC(q_t)}{2}$
- Profit max: **Quarter 30: MC_{30} = 1.93, a = 20.333**

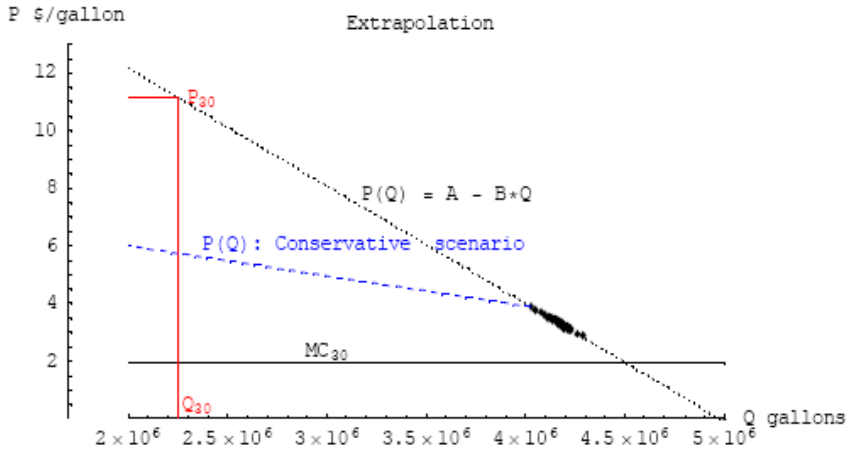
  $p_t = \frac{a}{2} + \frac{MC(q_t)}{2} = 11.3$; $q_t = 4,982,786 - 245,065p_t = 2,255,106$

Brand X: Concerns?

- $p(q) = 11.13$ is **<u>way</u>** outside our range of data
  - The relevant part of our estimated demand curve is based on HUGE extrapolation!
- Conservative scenario: What if the demand curve is very flat to the left of our observed data?
  - Then $p << 11.13$ would be optimal!
- However, even within the data range the highest prices are more profitable:

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Brand X: Caveat?



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Recap: Quantity Discounts (Segmentation)

- The optimal solution was interesting:
  - Alice gets 2 bars, and that is socially efficient! (MC =1 < than marginal WTP = 3)
  - Bob gets 1 bar, and that creates DWL! (MC =1 < than marginal WTP = 1.5)
- **Why?**
  - By making Bob's low quantity package smaller than what's efficient, it is not as good for him as the efficient size of 2 (he's 1.5 worse off gross)
  - But this makes Bob's low quantity package less desirable for Alice by 3 (gross)!
  - This in turn means that it's easier to squeeze more surplus out of Alice because her "cannibalizing" alternative (1-bar) is not very attractive for her.
- Cannibalization is a main concern in creating "packaging" (quantity discounts) because each consumer type has to have the incentive to buy what's tailored for them.

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Mercedes…

"You can make your Mercedes EV go faster for $60 a month"

1. The slower version is clearly inferior to the faster. How can an inferior version's availability improve Mercedes's profits?

This is called "versioning"

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Market Segmentation – Versioning

- The general ideas:
  - You can control the ***quality*** characteristics of a good:
    - **Professional and student software applications**
    - **High and low performance PCs**
    - **Business, economy, and basic economy**
  - High valuation customers are willing to pay more for quality upgrades
    - **Professionals and students**
    - **Gamers and poets**
    - **Business, leisure, and budget travelers**
- Is this a new type of economic problem that we have not yet seen?
- No! Think of low quality like 1 bar of chocolate and high quality like 2 bars…

Versioning – Examples

- Cars, Watches, Computers,…almost everything!
- Skip-the-line pricing at Ski resorts or Disney
- Airline fares:
  - Flexible ticketing (restricted vs. unrestricted fares)
  - First, business, Economy (Alice and Bob lesson 2: make the lowest quality version bad enough to reduce cannibalization risk from high WTP for quality buyers!)
- Emile Dupuit, a nineteenth century French economist:

  "It is not because of the few thousand francs which would have to be spent to put a roof over the third-class carriage or to upholster the third-class seats that some company or other has open carriages with wooden benches . . . What the company is trying to do is prevent the passengers who can pay the second-class fare from traveling third class; it hits the poor, not because it wants to hurt them, but to frighten the rich . . . And it is again for the same reason that the companies, having proved almost cruel to the third-class passengers and mean to the second-class ones, become lavish in dealing with first-class customers. Having refused the poor what is necessary, they give the rich what is superfluous." (Ekelund [1970])

Moving from Monopoly to Perfect Competition

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Competitive Markets

- What happens with many firms in a single market?
  - If there is slight differentiation then each firm has a more elastic demand due to the close substitute, but still has some price control.
  - If there is no differentiation, then the "law of one price" must obtain!
- **Perfectly Competitive Market.** Defined by:
  - Large number of sellers and buyers
  - A specific good where all goods produced are perfect substitutes
  - In the long run firm entry and exit occurs freely (no barriers to entry)

  $\Rightarrow$ Each firm (and consumer) is a ***price taker***

  (e.g., oil, metals, industrial plastic compounds, wood, etc.)

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The Price Taking Firm

- Conceptually, should the price taking firm make decisions in a different way compared to a monopolist?
- No!
  - The demand curve is just a horizontal line at some price p (Combs! Lecture 3)
  - Therefore, **MR = *p*** so we choose q^* at the point where MC = MR (= p)
  - Confirm that at q^* the firm covers its fixed costs and has positive profits

$\Rightarrow$ The MC curve is the firm's **supply curve** (in the region p > AC!)

- Can such a firm earn economic profits (above opportunity costs)?
  - Yes in the short run (though they can also run a loss in the short run)
  - In the long run possible, but unlikely (need a comparative cost advantage!)

Plywood…

"Plywood Becomes Hot Item in Housing Recovery" *WSJ*, March 21, 2013

1. Why has the price of plywood increased in 2013?
2. What is the relationship between the effect of an increase in demand for plywood on the price of plywood and the amount of time that sellers need in order to adjust to the increased demand?
3. What actions did lumber companies take to increase the quantity supplied of lumber, and what actions did they take in the past that has limited the current supply?

Competitive Market Supply

- Imagine a competitive market with many firms operating in it.
- **How much is produced in a price-taking market at price p?**
- Because each price-taking firm produces at a quantity that sets $\mathrm{MC} = p$ (which is just MC = MR), then for any given p the total amount produced will be the ***sum of individual firm productions***.
- Does this look familiar?
- This is just like what we did for aggregate demand!
  - Market supply is the "horizontal" aggregation of individual supplies (the MC curves of the firms)
- ***How are price and quantity determined in a competitive market?***

Home rent/prices…

"How an NYC Suburb Is Actually Managing to Bring Rents Down" *WSJ*, 8/5/25

1. From the article: "New Rochelle median rents are only 1.6% higher than in 2020, well below the 25% or higher increases in New York City and [other nearby]…markets…". Was the increase in demand greater or less than the increase in supply of homes in New Rochelle?
2. Why did rents in New York City increase so much more than rents in New Rochelle?

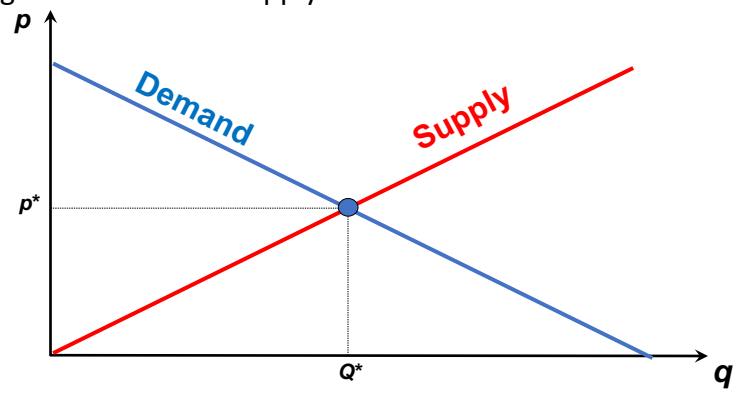
Car prices…

“Saving Money on a Used Car Is Getting Harder,” WSJ 8/12/25

1. How does the supply crunch of new cars during the pandemic impact the market for used cars today?
2. The article states that "Automakers have so far avoided across-the-board price hikes for new models despite Trump’s tariffs. But they are not expected to absorb multibillion-dollar hits to their bottom lines forever. Model year 2026 vehicles, going on sale soon, could see significant price increases." Would price increases in the market for new cars impact prices in the used car market? Why and how?

Competitive Market Equilibrium: Idea

- Graph aggregate demand and supply:



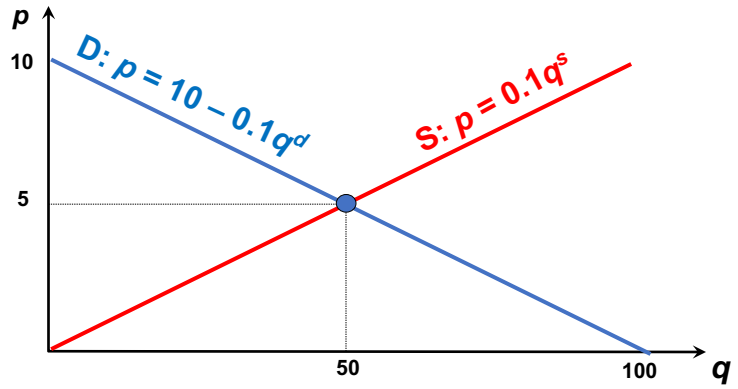
- Only at a p^* is supply = demand!
- Each firm produces at MC = p^* and the aggregate supply adds up to Q^*
- Each consumer consumes their optimal quantity, and all add up to Q^*

Competitive Market Equilibrium: Example

- Imagine 10 consumers (think many…)
  - Each with yearly demand for canola oil $q_i = 10 - p$,
  - so aggregate demand is $q^d = 100 - 10p$ (or $p = 10 - 0.1q^d$)
- Imagine 2 firms, (think many…) with yearly costs:
  - Low-cost firm 1: $C(q_1) = 30 + (q_1)^2/12 \Rightarrow MC_1 = q_1/6$
  - High-cost firm 2: $C(q_2) = 30 + (q_2)^2/8 \Rightarrow MC_2 = q_2/4$
- For any price p, each firm maximizes profits by producing at MC = p,
  - For firm 1: MC = $p \Rightarrow q_1/6 = p \Rightarrow q_1 = 6p$ (supply of firm 1)
  - For firm 2: MC = $p \Rightarrow q_2/4 = p \Rightarrow q_2 = 4p$ (supply of firm 2)
- So aggregate supply is: $q^s = 6p + 4p = 10p$ (or $p = 0.1q^s$)

Competitive Market Equilibrium: Example

- Graph aggregate demand and supply:



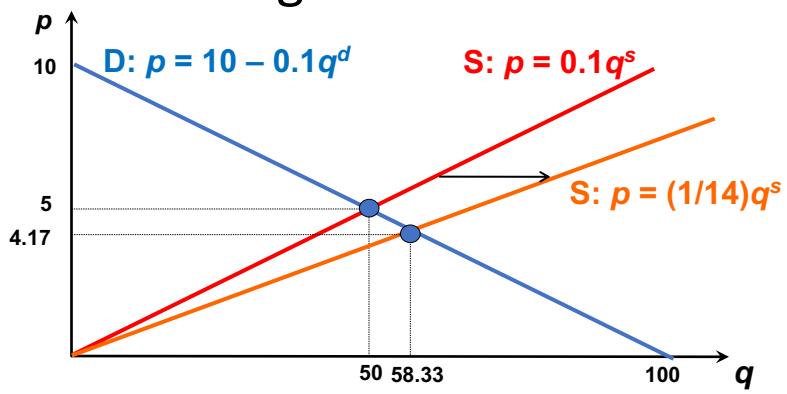
- Only at a p = 5 is supply = demand! ($10 - 0.1q = 0.1q \Rightarrow q = 50$)
  - $q_1 = 6p = 30$; $q_2 = 4p = 20$; $q_i = 10 - p = 5$ (each of the 10 buyers)
  - $\Pi_1 = 5\times30 - [30 + (30)^2/12] = 45$; $\Pi_2 = 5\times20 - [30 + (20)^2/8] = 20$

Short Run and Long Run

- Everything is great and both are making a profit. Is this the equilibrium?
  - Imagine that the low-cost firm-1 benefits form some trade secret that increases efficiency relative to the commonly known process used by the high-cost firm-2.
  - Imagine that anyone can start a company with firm 2's commonly known process
- Someone on the sidelines that is not making a profit may enter as firm 3!
  - Recall that for firm 1: MC = $p \Rightarrow q_1/6 = p \Rightarrow q_1 = 6p$
  - For firms 2 and 3: MC = $p \Rightarrow q_f/4 = p \Rightarrow q_f = 4p$ (*f* stands for 2 or 3, or "fringe")
- So aggregate supply is: $q^s = 6p + 4p + 4p = 14p$ (or $p = (1/14)q^s$)

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Short Run and Long Run



- At **$p = 4.17$** supply = demand! ($10 - 0.1q = (1/14)q \Rightarrow q = 58.33$)
  - $q_1 = 6p = 25$; $q_f = 4p = 16.66$; $q_i = 10 - p = 5.83$ (each buyer)
  - $\Pi_1 = 4.17 \times 25 - [30 + (25)^2/12] = 22.08$;
  - $\Pi_f = 4.17 \times 16.66 - [30 + (16.66)^2/8] = 4.72$

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Short Run and Long Run

- Everything is still great, and all firms are making a profit. Can another similar firm enter?
  - aggregate supply would be: $q^s = 6p + 4p + 4p + 4p = 18p$ (or $p = (1/18)q^s$)
  - Equate S = D to get $10 - 0.1q = (1/18)q \Rightarrow$ **$q = 64.29$ and $p = 3.57$**
  - $q_1 = 6p = 21.43$; $q_f = 4p = 14.28$; $q_i = 10 - p = 6.43$ (each buyer)
  - $\Pi_1 = 3.57 \times 21.43 - [30 + (21.43)^2/12] =$ **8.23**;
  - $\Pi_f = 3.57 \times 14.28 - [30 + (14.28)^2/8] =$ **-4.51** for three high-cost firms
- Would the third firm want to enter?
  - No! It would incur a loss (and so would everyone else)
- If it would enter what would happen?
  - Someone would choose to exit (depending on timing of fixed costs)

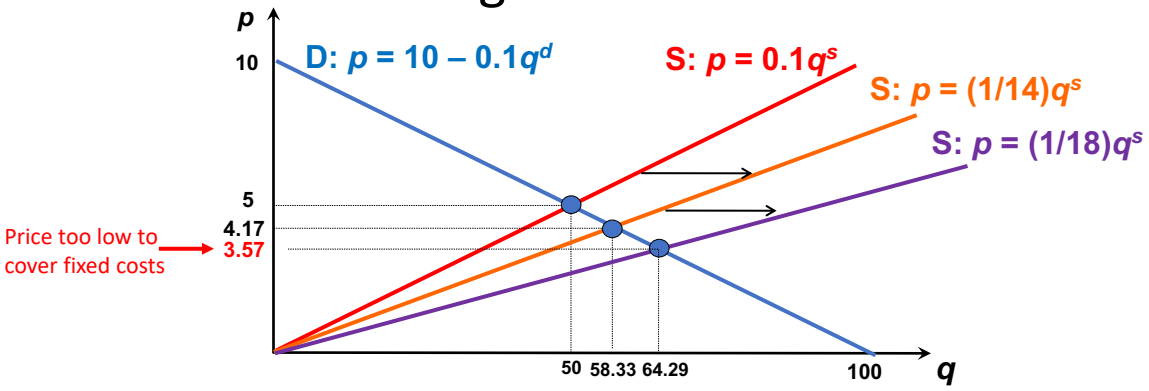
Competitive Market Equilibrium

- Recall the production possibilities:
  - Only one low-cost firm: $C(q_1) = 30 + q_1^2/12 \Rightarrow \text{MC}_1 = q_1/6$
    and $\text{MC}_1 = p$ implies that supply is $q_1 = 6p$
  - Possibly many fringe firms: $C(q_f) = 30 + q_f^2/8 \Rightarrow \text{MC}_f = q_f/4$
    and $\text{MC}_f = p$ implies that supply is $q_f = 4p$
- Hence, if there are n fringe firms then aggregate supply is

$$q^s = (6 + 4n)p \quad \text{or} \quad p = \frac{q^s}{6+4n}$$

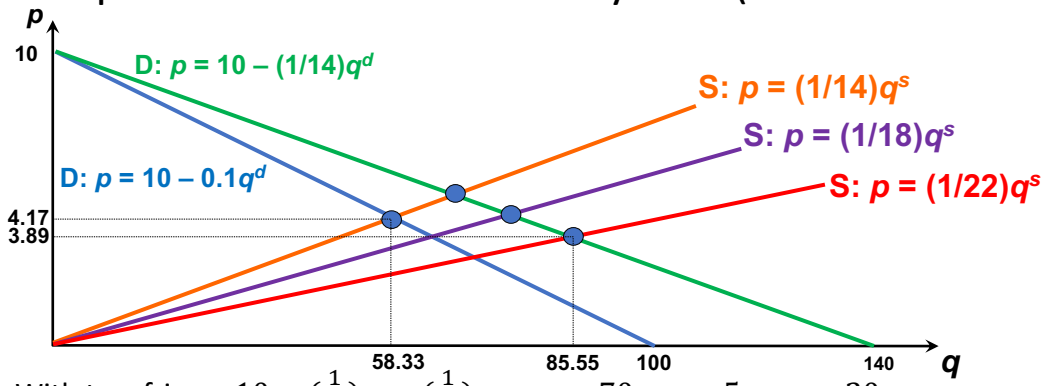
- We saw that with aggregate demand equal to $p = 10 - 0.1q^d$ (which was constructed by 10 consumers with $q_i = 10 - p$), the "long run equilibrium" had the low-cost firm and two fringe firms:

Short Run and Long Run



- With one fringe: $10 - q = 0.1q \Rightarrow q = 50,\ \ p = 5,\ \ \pi_f = 20$
- With two fringe: $10 - q = (\frac{1}{14})q \Rightarrow q = 58\frac{1}{3},\ \ p = 4.17,\ \ \pi_f = 4.72$
- With three fringe: $10 - q = (\frac{1}{18})q \Rightarrow q = 64.29,\ \ p = 3.57,\ \ \pi_f = -4.51$

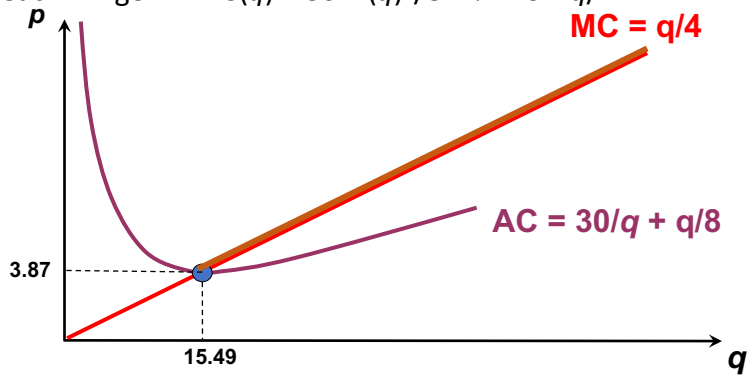
Example 2: Demand increases by 40% (more consumers)



- With two fringe: $10 - (\frac{1}{14})q = (\frac{1}{14})q \Rightarrow q = 70,\ \ p = 5,\ \ \pi_f = 20$
- With three fringe: $10 - (\frac{1}{14})q = (\frac{1}{18})q \Rightarrow q = 78.75,\ \ p = 4.375,\ \ \pi_f = 8.28$
- With four fringe: $10 - (\frac{1}{14})q = (\frac{1}{22})q \Rightarrow q = 85.55,\ \ p = 3.89,\ \ \pi_f = 0.26$
- ***Is this the "long run" supply?***

Short Run, Long Run and Costs

- Consider each fringe firm: $C(q) = 30 + (q)^2/8 \Rightarrow MC = q/4$



- Another interesting animal: $AC = 30/q + q/8$
- Where do they meet? $30/q+q/8 = q/4 \Rightarrow q = 15.49$, $p = 3.87$ (**min AC!**)
- How does this inform the firm's supply curve? *p = MC and p > AC* !!

Short Run, Long Run and Costs

- What conditions induce a firm to want to enter a competitive market?
  - Price is above AC! (make a profit)
  - Profits will cover the (sunk) cost of entry
- What conditions induce a firm to want to stay in a market?
  - Price is above AC! (make a profit), **OR…**
  - Fixed costs were sunk but variable costs are covered
- What conditions induce a firm to exit a market?
  - Price is below AC and fixed costs can be recovered

Short Run vs. Long Run Supply

- We started with one fringe firm and market supply equal to $q^s = 10p$, which would be the ***short-run market supply*** with those two firms.
- High profits, however, induced entry of a third firm to change the supply to $q^s = 14p$. Is this the long run market supply?
  - By definition, supply is $q^s(p)$, quantity supplied as a function of p.
  - But p is an **equilibrium construct** that depends on demand as well.
- If we think of long run as the response of supply to demand, then changes in demand will affect entry/exit!
  - Plywood?

Plywood 1/2006 – 10/2026



Summary: Short Run vs. Long Run Supply

- A fringe firm will be induced to enter if, after covering sunk entry costs, it will face a price greater than $3.87 (minimum AC)
- This implies that **long-run market supply**, if entry costs are low, will (almost) be a flat horizontal line at p = $3.87!
  - If demand "shifts up" (tastes, prices of substitutes rise, etc.) then short run prices will rise significantly above Min-AC, but this will induce entry that will suppress the price back towards Min-AC
  - If demand "shifts down" (tastes, prices of substitutes drop, etc.) then short run prices will fall below Min-AC, but this will induce exit by some "marginal" firm that will raise the price back towards Min-AC
- This is what the dynamics of a competitive market look like!

Competitive Markets: Summary

- **Perfect competition**: Many firms selling identical products are price takers—no individual firm controls the market price.
- **Firm decisions**: Each firm maximizes profits by producing where p = MC, provided operating is more profitable than shutting down.
- **Market equilibrium**: Aggregate supply is the sum of individual firms' supply curves; equilibrium occurs where market supply equals market demand.
- **Short-run dynamics**: Changes in demand can cause large price movements because existing firms have limited ability to adjust supply.
- **Long-run dynamics**: High profits attract entry, expanding supply and lowering prices; losses induce exit, reducing supply and raising prices.
- **Long-run competitive prices**: With free entry and similar production technologies, prices gravitate toward minimum average cost. Firms with lasting cost advantages can continue earning economic profits.